

Type I fiber decrease and ectopic fat accumulation in skeletal muscle from women with PCOS

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 2

December 4, 2023

Reviewed preprint version 1

May 26, 2023 (this version)

Sent for peer review

April 5, 2023

Posted to preprint server

March 8, 2023

E Stener-Victorin, G Eriksson, M Mohan Shrestha, C Perian, B Jude, V Engman, R Boi, E Nilsson, C Ling, J Nyström, I Wernstedt Asterholm, J T Lanner, A Benrick 

Department of Physiology and Pharmacology, Karolinska Institute, 17177 Stockholm, Sweden • Department of Physiology, Institute of Neuroscience and Physiology, Sahlgrenska Academy, University of Gothenburg, 40530 Gothenburg, Sweden • Epigenetics and Diabetes Unit, Department of Clinical Sciences, Lund University Diabetes Centre, Lund University, Malmö, Sweden • School of Health Sciences, University of Skövde, 54128 Skövde, Sweden

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Polycystic ovary syndrome's (PCOS) main feature is hyperandrogenism, which is linked to a higher risk of metabolic disorders in women. Gene expression analyses in adipose tissue and skeletal muscle reveal dysregulated metabolic pathways in women with PCOS, but these differences do not necessarily lead to changes in protein levels and biological function. To advance our understanding of the molecular alterations in PCOS, we performed global proteomic and phosphorylation site analysis using tandem mass spectrometry. Adipose tissue and skeletal muscle were collected at baseline from 10 women with and without PCOS, and in women with PCOS after 5 weeks of treatment with electrical stimulation. Perilipin-1, a protein that typically coats the surface of lipid droplets in adipocytes, was increased whereas proteins involved in muscle contraction and type I muscle fiber function were downregulated in PCOS muscle. Proteins in the thick and thin filaments had many altered phosphorylation sites, indicating differences in protein activity and function. The upregulated proteins in muscle post treatment were enriched in pathways involved in extracellular matrix organization and wound healing, which may reflect a protective adaptation to repeated contractions and tissue damage due to needling. A similar, albeit less pronounced, upregulation in extracellular matrix organization pathways was also seen in adipose tissue. Our results suggest that hyperandrogenic women with PCOS have higher levels of extramyocellular lipids, fewer oxidative and insulin-sensitive type I muscle fibers. These could be key factors leading insulin resistance in PCOS muscle while electric stimulation-induced tissue remodeling may be protective.

eLife assessment

This study offers an inventory of proteins and their phosphorylated sites that are up- and down-regulated in the adipose tissue and skeletal muscle of women with PCOS. The data were collected and analyzed using rigorous and validated methodology, making it a **useful** resource for identifying targets and strategies for future PCOS treatments. However, even though some of the predicted targets are **compelling**, further functional validation is required to ensure the accuracy of these identified targets. If confirmed, the findings of this study would be of significant interest to a wide range of readers.

Introduction

Polycystic ovary syndrome (PCOS) is a metabolic and endocrine disorder characterized by clinical signs of hyperandrogenism and reproductive dysfunction (Joham *et al.*, 2022 [DOI](#)). Although not part of the diagnosis, insulin resistance and abdominal obesity are common and lead to an increased risk of type 2 diabetes (Kakoly *et al.*, 2019 [DOI](#)). PCOS affects 8-17% of women worldwide. More than 50% of women with PCOS are overweight or obese, and obesity worsens all symptoms, including insulin resistance (Kakoly *et al.*, 2019 [DOI](#); Barber & Franks, 2021 [DOI](#)). In those with PCOS, obesity is associated with an altered adipose tissue function, increased visceral fat, adipocyte hypertrophy and lower adiponectin levels compared with BMI matched controls (Manneras-Holm *et al.*, 2011 [DOI](#); Villa & Pratley, 2011 [DOI](#); Manneras-Holm *et al.*, 2014 [DOI](#)). In recent independent bidirectional Mendelian studies, obesity is even thought to contribute to or cause the development of PCOS (Brower *et al.*, 2019 [DOI](#); Zhao *et al.*, 2020 [DOI](#); Liu *et al.*, 2022 [DOI](#)). Adipose tissue dysfunction also leads to consequences in other metabolic tissues. Overweight/obese women with PCOS are at increased risk of developing lipotoxicity due to increased triglyceride and free fatty acid levels and increased fatty acid uptake into non-adipose cells, including skeletal muscle, which is exacerbated by increased intra-abdominal fat with high lipolytic activity (Dumesic *et al.*, 2022 [DOI](#)). The lipotoxic state in skeletal muscle includes increased expression of genes involved in lipid storage and epigenetic changes, as demonstrated in a PCOS-like sheep model (Guo *et al.*, 2020 [DOI](#)). Consequently, excessive uptake and extracellular storage of free fatty acid in skeletal muscle promotes insulin resistance. Moreover, skeletal muscle from women with PCOS exhibit transcriptional, epigenetic and protein changes associated with insulin resistance, accompanied by an inflammatory, oxidative and lipotoxic state (Corbould *et al.*, 2005 [DOI](#); Skov *et al.*, 2007 [DOI](#); Nilsson *et al.*, 2018 [DOI](#); Stepto *et al.*, 2019 [DOI](#); Manti *et al.*, 2020 [DOI](#); Stepto *et al.*, 2020 [DOI](#); Moreno-Asso *et al.*, 2021 [DOI](#)).

Several studies show that defects in mRNA expression of mitochondrial function, fat oxidation and immunometabolic pathways in adipose tissue and skeletal muscle contribute to insulin resistance in women with PCOS (Skov *et al.*, 2007 [DOI](#); Nilsson *et al.*, 2018 [DOI](#); Stepto *et al.*, 2019 [DOI](#); Manti *et al.*, 2020 [DOI](#); Stepto *et al.*, 2020 [DOI](#)). Interestingly, the mechanism underlying the reduced insulin-stimulated glucose uptake in PCOS skeletal muscle likely differs from insulin resistance in BMI-matched controls (Corbould *et al.*, 2005 [DOI](#); Nilsson *et al.*, 2018 [DOI](#)). This finding is consistent with the theory that PCOS is a disorder with genetically distinct subtypes of PCOS (Dapas *et al.*, 2020 [DOI](#)). However, mRNA expression does not always translate to alterations in protein levels and changes in biological function. The first proteomics data on visceral fat and skeletal muscle from severely obese women with PCOS used mass spectrometry of selected protein spots on a 2D gel electrophoresis and were published a decade ago (Corton *et al.*, 2008 [DOI](#); Montes-Nieto *et al.*, 2013 [DOI](#); Insenser *et al.*, 2016 [DOI](#)). This method is limited as only a few proteins could be identified. Today, the combination of liquid chromatography with tandem mass spectrometry provides us with thousands of proteins. Recent publications map the proteome in serum, follicular fluid, ovary,

and endometrium from women with PCOS, and the proteome in serum may provide new biomarkers for PCOS (Li *et al.*, 2020 [↗](#); Abdulkhalikova *et al.*, 2022 [↗](#); Wang *et al.*, 2022 [↗](#)). We here use a nontargeted quantitative proteomics approach to advance our understanding of the pathophysiology in adipose tissue and skeletal muscle from women with PCOS.

We have previously shown that electrically stimulated muscle contractions, known as electroacupuncture, improve glucose regulation and decrease androgen levels in overweight/obese women with PCOS (Stener-Victorin *et al.*, 2016 [↗](#)). When acupuncture needles are stimulated by low-frequency electrical stimulation, they cause muscle contractions similar to those that occur during exercise. Muscle contractions activate specific physiological signaling pathways, and electrical muscle stimulations and exercise act through partially similar signaling pathways to induce glucose uptake in the acute response to muscle contractions (Benrick *et al.*, 2020 [↗](#)). In addition, long-term intervention with both exercise and electrical stimulation has been shown to improve glucose regulation, promote ovulation, decrease muscle sympathetic nerve activity and reduce circulating androgens in women with PCOS (Stener-Victorin *et al.*, 2009 [↗](#); Hutchison *et al.*, 2011 [↗](#); Jedel *et al.*, 2011 [↗](#); Johansson *et al.*, 2013 [↗](#); Stener-Victorin *et al.*, 2016 [↗](#); Tiwari *et al.*, 2019 [↗](#)). The molecular mechanisms mediating the effect in adipose tissue and skeletal muscle in response to long-term electrical stimulation, remain unclear. Therefore, the second aim is to use proteomics to provide mechanistic explanations for the improved glucose homeostasis in response to long-term electrical stimulation treatment.

Methods

Ethical approval

The study was conducted at the Sahlgrenska University Hospital and at the Sahlgrenska Academy, University of Gothenburg, Gothenburg, Sweden, in accordance with the standards set by the *Declaration of Helsinki*. Procedures have been approved by the Regional Ethical Review Board of the University of Gothenburg (approval number 520-11) and the study was registered at *ClinicalTrials.gov* [↗](#) (NCT01457209). All women provided oral and written informed consent prior to participation in the study.

Participants and study protocol

Women with PCOS were diagnosed according to the Rotterdam criteria (Group., 2004). This cohort has previously been described in detail and included 21 overweight and obese cases and 21 overweight and obese controls matched for age, weight and BMI (Kokosar *et al.*, 2016 [↗](#)). Of these, subcutaneous adipose tissue and skeletal muscle biopsies from 10 women with PCOS and 10 controls were included in the proteomics analysis. Participants reported to the laboratory in the morning after an overnight fast on menstrual cycle day 1–10 or irrespective of cycle day in amenorrheic/oligomenorrheic women. Anthropometrics were measured and BMI was calculated as previously described (Kokosar *et al.*, 2016 [↗](#)). Fasting blood samples were taken and HOMA-IR [(fasting insulin (mU/ml) x fasting glucose (mM)/22.5)] was calculated. Circulating testosterone was measured by GC-MS/MS (Kokosar *et al.*, 2016 [↗](#)). Anthropometrics, reproductive and endocrine variables for those included in the proteomics analysis are given in **Table 1** [↗](#).

Needle biopsies from subcutaneous adipose tissue from the umbilical area and skeletal muscle tissue from vastus lateralis were obtained under local anesthesia (Xylocaine, AstraZeneca AB, Södertälje, Sweden), from cases and controls (**Fig. 1A** [↗](#)). The fat biopsies were rinsed with saline before both tissues were snap frozen in liquid nitrogen and stored at -80°C until further analysis. Thereafter, women with PCOS received low-frequency electrical stimulations causing muscle contractions, so called electroacupuncture. Acupuncture needles were placed in somatic segments corresponding to the innervation of the ovaries and pancreas, bilaterally in abdominal muscle, in quadriceps muscles, and in the muscles below the medial side of the knee.

Variable	Controls baseline (n = 10)	PCOS baseline (n = 10)	Controls vs. PCOS (P-value)	PCOS treatment (n = 10)	Baseline vs. treatment (P-value)
Age (years)	31.2 ± 4.8	30.1 ± 5.1	0.60		
Weight (kg)	88.2 ± 11.0	88.6 ± 15.6	0.82	89.0 ± 14.9	0.33
BMI (kg/m ²)	30.5 ± 3.7	30.5 ± 4.1	0.88	30.7 ± 4.0	0.32
Ferriman Gallwey score	2.1 ± 1.7	7.7 ± 4.5	0.003	9.2 ± 6.0	0.48
Testosterone (pg/ml)	258 ± 77	476 ± 211	0.004	347 ± 220	0.012
Insulin (mU/ml)	8.2 ± 3.5	11.4 ± 8.5	0.47	8.5 ± 4.8	0.11
Glucose (mmol/L)	5.2 ± 0.3	4.8 ± 0.4	0.044	4.8 ± 0.3	0.67
HOMA-IR	1.92 ± 0.96	2.75 ± 2.00	0.41	1.89 ± 1.33	0.037
Hemoglobin A1c (mmol/mol)	31.5 ± 2.6	31.7 ± 2.7	1.00	30.5 ± 2.9	0.035
Triglycerides (mmol/L)	0.71 ± 0.18	1.11 ± 0.58	0.076	1.01 ± 0.58	0.066

Data are presented as mean ± SD. Differences between PCOS and controls were analyzed by Mann-Whitney *U* test. Wilcoxon signed-rank test was used to analyze changes between measurements at baseline and after 5 weeks of treatment.

Table 1.

Anthropometric and biochemical analyses in study participants.

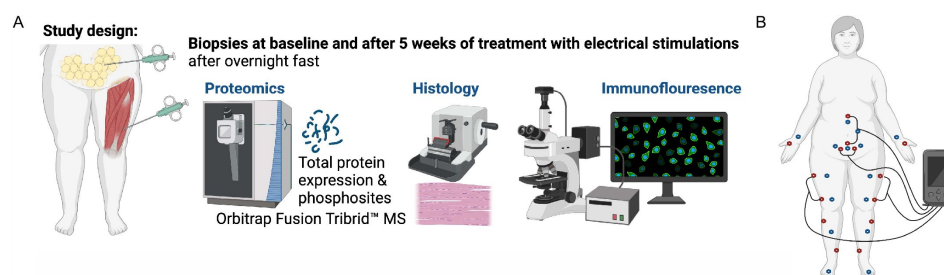


Fig. 1.

A) Muscle and fat biopsies collected from 10 controls and 10 women with PCOS at baseline and after treatment with electrical stimulations. Electrical stimulations were given 3 times/week for 5 weeks. B) The electrical stimulation protocol alternating between protocol 1 in red dots and protocol 2 in blue dots. Acupuncture points not connected to the stimulator were stimulated manually. Created with *BioRender.com*

Needles were inserted to a depth of 15–40 mm with the aim of reaching the muscles. Needles were connected to an electrical stimulator (CEFAR ACUS 4; Cefar-Compex Scandinavia, Landsbro, Sweden) and stimulated with a low-frequency (2 Hz) electrical signal for 30 minutes. The intensity was adjusted every 10th minute due to receptor adaptation, with the intention to produce local muscle contractions without pain or discomfort. Treatment was given three times per week over 5 weeks, and the number of treatments varied from 11 to 19. Two sets of needle placements were alternated to avoid soreness (**Fig. 1B**) (Stener-Victorin *et al.*, 2016). Baseline measurements were repeated after the five weeks of treatment, within 48 hours after last treatment, and new fat and muscle biopsies were collected. After all relevant clinical information was obtained, samples were coded and anonymized.

Proteomic sample preparation and LC-MS/MS analysis

Detailed method information is given in **Supplemental data 1**. In short, muscle and adipose tissue biopsies (20–50 mg) were homogenized and the protein concentrations were determined using Pierce™ BCA Protein Assay Kit (Thermo Fischer Scientific). Reduced samples were then processed using the modified filter-aided sample preparation (FASP) method (Wisniewski JR *et al.* Nat Methods. 2009 May;6(5):359–62) with the addition of two additional washes with n butanol:methanol (3:1). Pierce trypsin protease (MS Grade, Thermo Fisher Scientific) was added to generate cleaved peptides, which were labelled using Tandem Mass Tag (TMT pro 16plex) reagents according to the manufacturer's instructions (Thermo Fischer Scientific). For the total proteome analysis, aliquots were fractionated into 40 primary fractions by basic reversed-phase chromatography (bRP-LC) using a Dionex Ultimate 3000 UPLC system (Thermo Fischer Scientific). The primary fractions were concatenated into final 20 fractions (1+21, 2+22, ... 20+40), evaporated and reconstituted. The enriched phosphopeptide samples were fractionated into 20 primary fractions on the same LC setup, and the primary fractions were concatenated into final 10 fractions (1+11, 2+12, ... 10+20), evaporated and reconstituted. The fractions were analyzed by LC-MS/MS on an Orbitrap Fusion Tribrid™ mass spectrometer interfaced with an Easy-nLC 1200 liquid chromatography system (both Thermo Fisher Scientific).

Proteomic data analysis

Identification was performed using Proteome Discoverer version 2.4 (Thermo Fisher Scientific). The database search was performed using the Mascot search engine v. 2.5.1 (Matrix Science, London, UK) against the Swiss-Prot Homo sapiens database. For phosphopeptide samples, phosphorylation on serine, threonine and tyrosine was added as a variable modification. Quantification was performed in Proteome Discoverer 2.4. TMT reporter ions were identified with 3 mmu mass tolerance in the MS3 HCD spectra for the total proteome experiment and with 20 ppm mass tolerance in the MS2 HCD spectra for the phosphopeptide experiment, and the TMT reporter S/N values for each sample were normalized within Proteome Discoverer 2.4 on the total peptide amount. Only the unique identified peptides were considered for protein quantification. The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the Proteomics Identifications (PRIDE) (RRID:SCR_003411) (Deutsch *et al.*, 2020) partner repository with the dataset identifier PXD025358.

The normalized abundance counts were used for the downstream analysis using the Differential Enrichment analysis of Proteomics data package (DEP) (Zhang *et al.*, 2018). The counts were log₂ transformed and proteins that were quantified in less than 2/3 of the samples were removed. Missing values were imputed using random draws from a Gaussian distribution around the minimal value as the missing value was not random but concentrated to proteins with low intensities. The data was batch effect adjusted using ComBat (Johnson *et al.*, 2007) with the LC-MS/MS run assigned as the batch covariate. Differential expression analysis was performed on the dataset with DEP's test_diff function which uses protein-wise linear models and empirical Bayes statistics using limma (Smyth, 2004). The differential expression analysis calculated the log₂ fold-changes (FC), *p*-values and *q*-values between the three groups control and PCOS at baseline

(W0), and PCOS after 5 weeks of treatment (W5) generating two different result datasets. Proteins and phosphorylation enrichment were determined to be significantly differentially expressed between the groups if the *p*-value < 0.05, and log2 FC ≥ 0.5 or log2 FC ≤ -0.5.

mRNA expression and DNA methylation arrays

Detailed method information is given in **Supplemental data 1**. A HumanHT-12 v4 Expression BeadChip array (Illumina) was used to analyze global mRNA expression and the Infinium HumanMethylation450k BeadChip array (Illumina) was used to analyze genome-wide DNA methylation in adipose tissue and skeletal muscle biopsies from those with PCOS taken before and after 5 weeks of electrical stimulation. The bioinformatics analyses of DNA methylation array data were performed as described previously (Ronn *et al.*, 2015 [\[1\]](#)).

Pathway enrichment analysis

The tool Uniprot, Universal Protein Resource (RRID:SCR_002380), was used to retrieve protein names. We applied enrichment analysis to all differently expressed proteins and phosphorylation sites using Enrichr (RRID:SCR_001575) and STRING (RRID:SCR_005223). Ontology terms with a *q*-value < 0.05 and including at least 3 proteins/phosphosites or an odds ratio > 100 were considered as enriched.

Histological analyses and immunofluorescence

Skeletal muscle and adipose tissue biopsies were fixed in histofix (Histolab, Sweden) for >72h and then stored in 70% ethanol. Tissues were dehydrated and fixated in paraffin blocks. Paraffin-embedded adipose tissue and muscle tissue (7 μm) were cut into sections using a rotary microtome (Leica Microtome) and mounted on Superfrost™ Plus Adhesion microscope glass slides (Epredia™ J1800AMNZ, #10149870, Thermo Fisher Scientific). Picrosirius Red staining (cat#24901-250, Polysciences, Inc.) was used to identify fibrillar collagen in adipose and muscle tissue. Pictures (4-5 sections/subject) were taken at 10x, and 20x magnification using a light microscope (Olympus BX60 & PlanApo, 20x/0.7, Olympus, Japan). All images were analyzed on ImageJ software v1.47 (National Institutes of Health, Bethesda, MD, USA).

For immunofluorescence, the muscle sections were deparaffinized twice in xylene (#534056, Sigma-Aldrich) for 5 min. Sections were rehydrated stepwise twice in 100% ethanol and once in 95%, 70%, 50% ethanol and in deionized water for 5 min each before a final rinse in PBS (#18912-014, Gibco, pH 7.4) for 5 min. The slides were subjected to heat-induced antigen retrieval by heating in antigen retrieval buffer (10 mM Citric acid monohydrate, 0.05 % (v/v) Tween-20, pH 6.0) until it reaches the boiling point and cooling to room temperature. The tissue sections were incubated in blocking buffer (3% normal donkey serum (NDS) (v/v) in PBS) at room temperature for 1 h followed by overnight incubation at 4 °C with primary antibodies (rabbit anti-perilipin-1, dilution 1:150 (Abcam Cat# ab3526) and mouse anti-myosin (skeletal slow), dilution 1:300 (MYH7 antibody, Sigma-Aldrich Cat# M8421) diluted in incubation buffer (PBS containing 0.3% Triton X-100, 1% BSA, 1% NDS and 0.01% sodium azide, pH 7.2). After rinsing the slides three times for 10 min each in PBS, the sections were incubated with fluorochrome-conjugated secondary antibodies (Donkey anti-Rabbit IgG (H+L) Highly Cross-Absorbed Secondary Antibody, Alexa Fluor™ 555, diluted 1:250 (Thermo Fisher Scientific Cat# A-31572) and Donkey anti-Mouse IgG (H+L) Highly Cross-Absorbed Secondary Antibody, Alexa Fluor™ Plus 448, diluted 1:250 (Thermo Fisher Scientific Cat# A32766) diluted in incubation buffer. The muscle sections were rinsed three times for 10 min each in PBS. The slides were mounted with coverslips (#ECN 631-1574, VWR) using Vectashield antifade mounting medium with DAPI (#H-1200, Vectashield). Images were obtained using a Zeiss LSM 700, AxioObserver microscope with Plan-Apochromat 10x/0.45 M27 objective lens. Argon lasers of 488 nm and 555 nm wavelengths were used to excite Alexa Fluor 488 (green) and Alexa Fluor 555 (red) respectively, and Laser Diode 405 nm to excite DAPI (blue). Skeletal muscle fiber size and type were quantified in muscle biopsies frozen in Tissue-Tek® O.C.T. Compound (Sakura

Finetek, Gothenburg, Sweden). Cross sections (10 μ m) were cryosectioned using a NX70-Epredia cryostat, moved onto glass slides (Expredia, J1800AMNZ) and stored at -20°C. The sections were subsequently immunohistochemically stained for type I fibers and fiber boundaries. In brief, the sections were dried in room temperature for 60 minutes and then fixed in 4% formaldehyde (Merck, 100496) for 30 minutes, permeabilized with 0.5% PBS-Triton™ X-100 (Sigma-Aldrich, 9036-19-5) for 20 minutes and thereafter incubated with 0.25% PBS-Triton™ X-100 with 10% goat serum for 30 minutes. The sections were then incubated with primary MYH7 antibody (1:25; DSHB, BA-F8) for type I fibers overnight at 4°C and subsequently secondary antibody Alexa Flour™ 568 (1:500; Thermo Fisher, A-11031) for 60 minutes at room temperature, both in 0.1% PBS-Triton™ X-100 with 1% BSA. Finally, the sections were incubated with WGA Oregon Green™ 488 (Invitrogen, W7024) for fiber boundaries for 3 hours, whereafter Fluoromount-G™ mounting medium (Thermo Fisher, 00-4959-52) and cover slips were applied. The slides were visualized using a Zeiss AxioScan.Z1 slide scanner. Fiber cross sectional area were automatically determined using MyoVision v1.0 and the proportion of type I fibers were manually counted on ImageJ.

Statistics

Differences in clinical characteristics between women with PCOS and controls were assessed using the Mann-Whitney U test and data are presented as mean $\pm$ SD. Wilcoxon signed-rank test was used to analyze changes between measurements at baseline and after 5 weeks of treatment. Differences in protein expression were calculated on log2 fold-change. Proteins and phosphorylation enrichments were determined to be significantly differentially expressed between cases and controls, and after 5 weeks of treatment in women with PCOS, if the *p*-value < 0.05, and log2 FC $\geq$ 0.5 or log2 FC $\leq$ -0.5. Proteomic data are presented as log2 fold-change.

Results

Clinical characteristics

Women with PCOS had more antral follicles < 9 mm (22.7 ± 7.9 vs 9.4 ± 4.1 , *P*=0.001), larger ovary volume (8.0 ± 2.9 vs 5.0 ± 2.7 ml, *P*=0.028), higher Ferriman-Gallwey score, and higher circulating testosterone than controls (**Table 1**). Six of the 10 women with PCOS met all three PCOS criteria; two had hyperandrogenemia and PCO morphology, one had hyperandrogenism and irregular cycles, and one had irregular cycles and PCO morphology. Five weeks of treatment lowered testosterone, HOMA-IR, and HbA1c levels, tended to decrease triglyceride levels but did not improve Ferriman-Gallwey score (**Table 1**).

Total protein expression and phosphorylation in skeletal muscle

In total, we identified 3480 proteins in skeletal muscle. 58 unique proteins were found to be differentially expressed in skeletal muscle from women with PCOS versus controls (*p* < 0.05, and log2 FC $\geq$ 0.5 or $\leq$ -0.5, **Fig. 2A**, **Table S1**). 25 proteins were upregulated and 33 were downregulated in women with PCOS and the expression log2 fold change in expression ranged from -3.06 to 1.21 (**Table S1**). We searched for enriched signaling pathways among the differently expressed proteins using STRING analysis. Our network had significantly more interactions than expected (enrichment *p*-value < $1e^{-16}$). This means that the differently expressed proteins have more interactions with each other than what would be expected from a random set of proteins of the same size. Such an enrichment indicates that the proteins are biologically linked. Upregulated proteins were enriched in lipid metabolic pathways including negative regulation of cholesterol transport, regulation of lipoprotein lipase activity, and negative regulation of metabolic processes (**Table S2**). This enrichment was driven by increased expression of apolipoprotein C-I, C-II and C-III (**Fig. 2B**), which are also enriched in the negative regulation of lipoprotein lipase activity (GO:0051005). Aldo-keto reductase family 1 members C1 and C3 (*AKR1C1* and *AKR1C3*, **Fig. 2B**), which have an androsterone dehydrogenase activity (GO:0047023), were also upregulated

suggesting that muscle may produce dihydrotestosterone via the backdoor pathway. Moreover, perilipin-1 that typically coats the surface of lipid droplets in adipocytes (Gandolfi *et al.*, 2011; Zhao *et al.*, 2021), so called extra-myocellular adipocytes, was increased in PCOS muscle. The increased expression of perilipin-1 was confirmed by immunofluorescence staining of muscle biopsies (Fig. 2C).

The down-regulated proteins in PCOS were enriched in pathways involved in muscle contraction, muscle filament sliding, and transition between fast and slow fibers (Fig. 3A). All significantly enriched pathways are listed in Table S2. Expression of myosin heavy chain beta, which is specific for type I muscle fibers was decreased in PCOS (Fig. 3B). Several proteins that are more highly expressed in type I muscle fibers, consistently had a lower expression in women with PCOS, for example myosin 7, myosin regulatory light chain 2 and 3, troponin I and troponin T in slow skeletal muscle, and the Ca^{2+} pump SERCA2a. These are proteins located in both the thick filaments (myosin regulatory light chains) and the thin filaments (troponins) of slow-twitch fibers (Fig. 3C). A decrease in type I slow-twitch muscle fibers was also supported by staining of human myosin heavy chain (MYH7) as a marker (Fig. 3D). To further assess whether there was a reduced fiber size or decreased number of type I fibers; fiber cross sectional area of the fibers and the percentage of type I fibers were analyzed. The quality of muscle biopsies was impaired in the PCOS group; therefore 60% fewer fibers from each individual were analyzed in the PCOS group compared with controls ($P=0.02$). There was no significant difference in the mean cross-sectional area of the fibers (4530 ± 720 in controls versus $4281 \pm 902 \mu\text{m}$ in PCOS, $P=0.64$, $n=5-7/\text{group}$) or the percentage of type I fibers (48 ± 12 versus $45 \pm 17 \mu\text{m}$, $P=0.69$, $n=4/\text{group}$) in the relatively few individuals analyzed (Fig. 3E). There was however a decrease in the individual fiber cross-sectional area in PCOS muscle versus controls (Fig. 3F).

We searched for overlap between the differentially expressed proteins in skeletal muscle in this study and the differentially expressed genes in our previous meta-analysis of gene expression array data (Manti *et al.*, 2020). As suspected, the overlap between gene expression and protein levels was small. We found that one upregulated and 12 downregulated genes in muscle biopsies from women with PCOS were also differently expressed at the protein level in this study (Table S3). Several proteins involved in skeletal muscle contraction were consistently downregulated at the mRNA expression level in muscle tissue from women with PCOS including *MYL3*, *MYOZ2*, *TNNT1*, *LMOD2*, *NRAP*, and *XIRP1* (Table S3).

We identified 5512 phosphosites in muscle, and 61 sites in 40 unique proteins were differentially phosphorylated in PCOS versus controls (Table S4), suggesting different protein activity. Eleven of the differently expressed proteins had one or more differently phosphorylated sites, including increased phosphorylation of Ser 130, 382, and 497 in perilipin-1. Many of the proteins in the thick and thin filaments had one or more altered phosphorylation sites (Fig. 3G). There were no significantly enriched pathways among the differently expressed phosphosites.

Total protein expression and phosphorylation in adipose tissue

In total, we identified 5000 proteins in adipose tissue but the difference between groups was modest. 21 unique proteins were found to be differentially expressed in adipose tissue from women with PCOS versus controls ($p < 0.05$, and $\log_2 \text{FC} \geq 0.5$ or ≤ -0.5 , Fig. 4A). Six proteins were upregulated and 15 were downregulated in women with PCOS and the expression $\log_2$ fold-change in expression ranged from 2.1 to -1.6 (Fig. 4B, Table S5). Several of the upregulated proteins play a role in immune system processes including, immunoglobulins, HLA class I histocompatibility antigen, and sequestosome 1. Sequestosome 1 may also regulate mitochondrial organization (Poon *et al.*, 2021). Three mitochondrial matrix proteins; tRNA pseudouridine synthase A, Enoyl-CoA hydratase domain-containing protein 2 and NAD kinase 2 had an altered expression (Fig. 4B), possibly indicating mitochondrial dysfunction. There were no significantly enriched signaling pathways in adipose tissue.

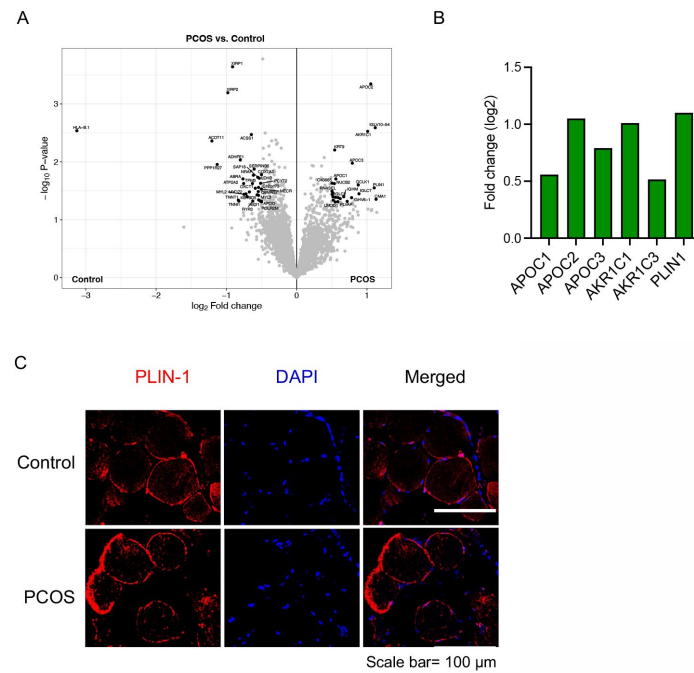


Fig. 2.

A) Volcano plot showing the mean protein log₂ fold change in skeletal muscle (PCOS vs. Control) plotted against the -log₁₀ *p* value highlighting significantly regulated proteins in black (*p* < 0.05, log₂ fold change ± 0.5). *n* = 10/group. B) Increased protein expression of apolipoproteins, aldo-keto reductase AKR1 C1 and C3, and perilipin-1 in those with PCOS. C) Staining of perilipin-1 and DAPI in skeletal muscle.

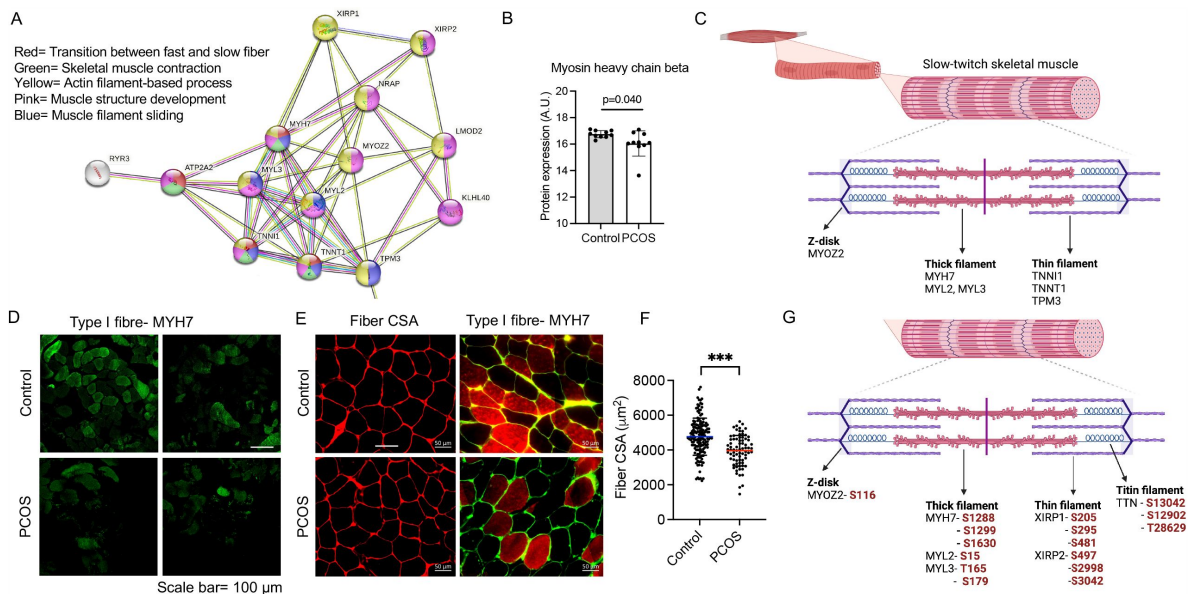


Fig. 3.

A) Protein network using STRING analysis on proteins with a lower expression in PCOS skeletal muscle vs. controls. Lines indicate protein-protein associations. B) Decreased expression of the type I fiber specific protein myosin heavy chain beta (MYH7) in those with PCOS. C) Lower expression of proteins in slow-twitch type I muscle fibers in PCOS vs. controls (*p* < 0.05, log₂ fold change < -0.5). D, E) Immunofluorescent staining of type I muscle fibers with myosin heavy chain beta, and E) the cell membrane with WGA. F) Quantification of fiber cross-sectional area (CSA) in E). G) Differently phosphorylated sites in proteins expressed in muscle filaments (*p* < 0.05, log₂ fold change ± 0.5).

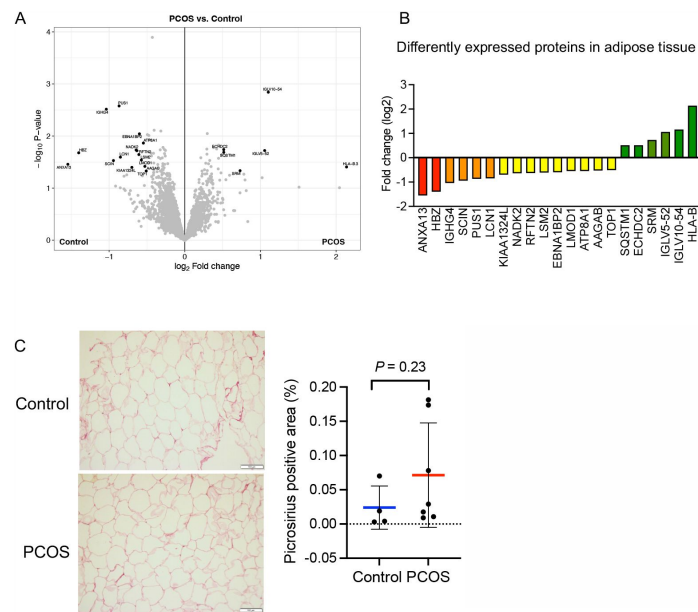


Fig. 4.

A) Volcano plot showing the mean protein log₂ fold change in adipose tissue (PCOS vs controls) plotted against the -log₁₀ p value highlighting significantly regulated proteins (black; p<0.05, log₂ fold change +/-0.5). n=10/group. B) All differentially expressed proteins in adipose tissue from women with PCOS. C) Picrosirius red staining of s.c. adipose tissue. Difference between women with PCOS and controls was based on Mann-Whitney U test and is presented as mean ± SD.

Both low-grade inflammation and transforming growth factor beta (TGF β)-induced fibrosis have been suggested to play a role in the pathophysiology of PCOS (Mancini *et al.*, 2021 [↗](#); McIlvenna *et al.*, 2021 [↗](#)). Dysregulated signaling of TGF β has been linked to the development of ovarian fibrosis and reproductive dysfunction (Stepto *et al.*, 2019 [↗](#); McIlvenna *et al.*, 2021 [↗](#)). Women with PCOS have elevated levels of circulating TGF β 1 (Raja-Khan *et al.*, 2010 [↗](#)), which is thought to trigger increased fibrotic mechanisms in other peripheral tissues. However, we did not detect an increased abundance of fibrous collagens in adipose tissue and the fibrillar collagen levels as judged by Picrosirius red staining were similar between groups although the variability was higher in the PCOS group (**Fig. 4C** [↗](#)).

We identified 5734 phosphosites in adipose tissue, of which 39 were differently phosphorylated sites in 34 unique proteins. Ten of these sites had lower phosphorylation (**Table S6**). There was no overlap between differently phosphorylated proteins and differently expressed proteins. Perilipin-1 had two phosphorylation sites with higher phosphorylation; Ser 497 and 516 in PCOS adipose tissue compared to controls (**Table S6**). Ser 497 showed increased phosphorylation in perilipin-1 in both muscle and adipose tissue. Under adrenergic stimulation, perilipin-1 is phosphorylated at Ser 497 by protein kinase A, which in turn triggers lipolysis by hormone sensitive lipase (HSL) in adipocytes (Marcinkiewicz *et al.*, 2006 [↗](#)).

Skeletal muscle protein expression and phosphorylation changes in skeletal muscle after treatment with electrical stimulation

Since long-term electrically stimulated muscle contractions improve glucose regulation and lower androgen levels in women with PCOS (Stener-Victorin *et al.*, 2016 [↗](#)), we analyzed genome-wide mRNA expression from women with PCOS after treatment to identify changes in skeletal muscle gene expression in response to stimulation. None of the transcripts exhibited changes in expression after FDR correction after 5 weeks of treatment ($q < 0.05$), but 12 transcripts had a FC > 1.2 ($p < 0.05$, **Table S7**), of which 5 were different collagens. We also analyzed whether the response to electrical stimulation was associated with DNA methylation changes in skeletal muscle. We found that 41,186 (13.8%) of 298,332 analyzed CpG sites had differential methylation in skeletal muscle after treatment ($p < 0.05$), which is almost three times more than expected by chance ($p < 0.0001$, Chi2 test). Of these, 43 CpG sites remained significant after FDR correction ($q < 0.05$, **Table S7**). The majority of the sites (74%) showed decreased methylation in response to treatment. The absolute change in DNA methylation in response to treatment ranged from 3 to 14 percentage points.

Since mRNA expression was not significantly regulated in response to repeated electrical stimulations, we investigated whether the effects were regulated at the protein level. We found that 376 unique proteins were changed in skeletal muscle after treatment with electrical stimulation ($p < 0.05$, **Fig. 5A** [↗](#), **Table S8**). Most proteins were upregulated in women with PCOS after treatment (98%), and the expression log2 fold-change in expression ranged from -1.37 to 2.15. The upregulated proteins were enriched in signaling pathways involved in extracellular matrix (ECM) organization, regulation of TGF β production, neutrophil-mediated immunity, wound healing, and blood coagulation (**Fig. 5B** [↗](#), **Table S9**). Proteins involved in ECM organization include 8 different collagens, integrins and TGF- β 1 (**Table S9**). Collagen 1A1, 1A2 and VCAN were increased after treatment at both gene and protein levels, implying that acupuncture needling elicits a wound healing response. There was a trend towards increased staining of fibrous collagen after treatment, but this was not significant, potentially due to the low number of analyzed good quality sections (**Fig. 5C** [↗](#)). Other upregulated signaling pathways include exocytosis and vesicle transport along the actin filament, muscle contraction and actin filament organization and negative regulation of adenylate cyclase-activating adrenergic signaling. Several effects previously shown to be upregulated after one bout of electroacupuncture were regulated in the opposite direction as pathways involved in negative regulation of angiogenesis, negative regulation of

blood vessel morphogenesis, negative regulation of nitric oxide metabolic processes, and regulation of vasoconstriction were enriched. None of the 58 proteins that had a different expression in the PCOS group at baseline were reversed after 5 weeks of treatment.

198 phosphosites in 152 unique proteins showed a changed phosphorylation in response to electrical stimulation. 178 sites had higher phosphorylation, and 46 sites were less phosphorylated (**Table S10**). There were no significant enriched pathways among the proteins with differently expressed phosphorylation sites. 38 of the differently expressed proteins had one or more differently phosphorylated sites. These proteins, with changes in both total protein and phosphorylation levels, were enriched in actin filament organization (GO:0007015).

Total protein expression and phosphorylation in adipose tissue after treatment

Similar to skeletal muscle, long-term electrical stimulations had minimal effects on gene expression in adipose tissue. None of the transcripts exhibited changes in expression after FDR correction ($q < 0.05$) after 5 weeks of treatment, or had a FC > 1.2 (data not shown). We found that 23,517 (7.9%) of 298,289 analyzed CpG sites had differential methylation in adipose tissue after 5 weeks of treatment ($p < 0.05$), which is more than expected by chance ($p < 0.0001$, Chi2 test). The majority (63.5%) of these sites showed reduced methylation in response to treatment. One CpG site remained significant after FDR correction ($q < 0.05$, -2.2% points reduced methylation of cg13383058 in the transcription start site of *CD248*). Therefore, we investigated whether the long-term effects were regulated at the protein level. 61 unique proteins were found to be changed in adipose tissue after electrical stimulation treatment (**Fig. 6A**, **Table S11**). Most of the proteins were upregulated (85%) and 9 were downregulated in women with PCOS after treatment, and the expression log2 fold-change in expression ranged from -0.89 to 1.41. The upregulated enriched signaling pathways include ECM organization and Fc-gamma receptor signaling (**Table S12**).

In accordance with these findings, 5 weeks of treatment increased the fibrillar collagen content in adipose tissue (**Fig. 6B**). The expression of DNA topoisomerase 1 and leimodulin-1 was lower in women with PCOS than in controls but increased after treatment (**Table S11**). Leimodulin-1 is required for proper contractility of smooth muscle cells by mediating nucleation of actin filaments, and myosin regulatory light polypeptide 9 plays an important role in regulating smooth muscle contractile activity. The enzyme prostacyclin synthase, a potent mediator of vasodilation, was also increased and could act on the smooth muscles in the vessel wall. 49 phosphosites in 46 unique proteins showed altered phosphorylation in response to electrical stimulation. All except 4 sites had higher phosphorylation (**Table S13**). There were no significantly enriched pathways among the proteins with differently expressed phosphosites.

11 proteins showed higher expression in both skeletal muscle and adipose tissue after treatment; FCGR3A, FAP, PTGIS, COL1A2, COL14A1, COL1A1, MYL9, LMNB1, SIRPA, ARPC2, and RAP2A. These proteins are enriched in ECM organization (GO:0030198), negative regulation of nitric oxide biosynthetic/metabolic processes (GO:0045019, GO:1904406), and Fc-gamma receptor signaling pathways (GO:0038096, GO:0038094, GO:0002431).

Discussion

Proteome signature in PCOS

We have profiled the proteome of skeletal muscle and adipose tissue to advance our understanding of the pathophysiology of PCOS. The changes in protein expression in adipose tissue were small, whereas in skeletal muscle of women with PCOS there was a clear downregulation of proteins involved in muscle contraction. The skeletal muscle contains a

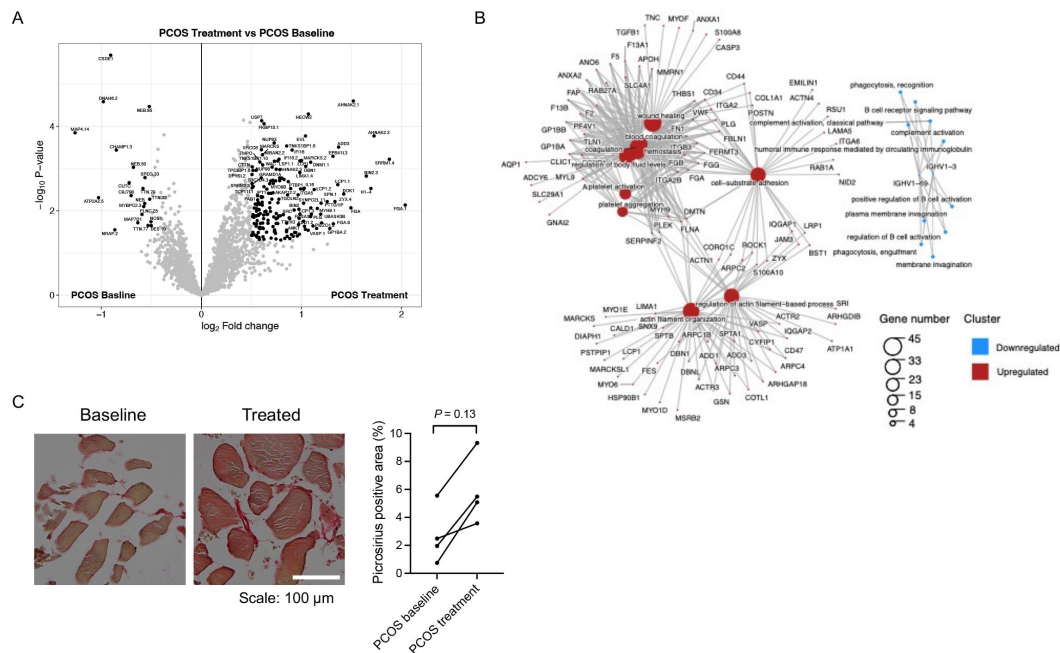


Fig. 5.

A) Volcano plot showing the mean protein log₂ fold change in skeletal muscle (treatment vs baseline in PCOS) plotted against the -log₁₀ p value highlighting significantly regulated proteins (black; p<0.05, log₂ fold change +/-.5). n=10/group. B) GO terms for biological function of the changed proteins. C) Representative pictures and quantification of picrosirius red staining of skeletal muscle before and after treatment with electrical stimulation in the same individual. Changes between baseline and after treatment was based on Wilcoxon signed-rank test.

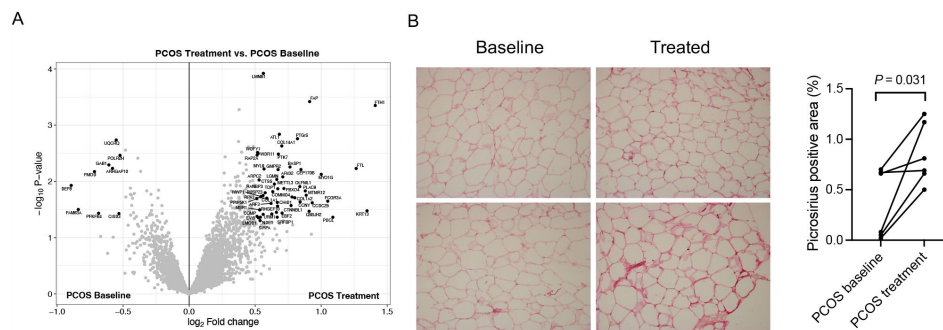


Fig. 6.

A) Volcano plot showing the mean protein log₂ fold change in adipose tissue (treatment vs baseline in PCOS) plotted against the -log₁₀ p value highlighting significantly regulated proteins (black; p<0.05, log₂ fold change +/-.5). n=10/group. B) Representative pictures and quantification of picrosirius red staining of adipose tissue before and after treatment with electrical stimulation. Changes between baseline and after treatment was based on Wilcoxon signed-rank test.

mixture of slow-twitch oxidative and fast-twitch glycolytic myofibers, which exhibit different physiological properties. Type I, or red fibers, are slow-twitch fatigue resistant muscle fibers which have higher mitochondrial and myoglobin content, and thus more aerobic than type II fast-twitch fibers. Several proteins specific to or known to be highly expressed in type I muscle fibers, were consistently downregulated in women with PCOS. These proteins are located in both the thick and the thin filaments of slow-twitch fibers. These data suggest that there are fewer type I fibers, decreased number and/or smaller, in the PCOS muscle. Unfortunately, we were unable to quantify the number of type I fibers in the entire cohort because of the poor quality of the PCOS muscle biopsies and the unavailability of muscle tissue. We also identified several differently phosphorylated sites in proteins located in the thick and thin filaments, indicating a differential protein activity, since phosphorylation can change the activity of a protein.

Here we show that the signaling pathway important for the transition between fast and slow fibers was downregulated and that individuals with PCOS had lower expression of myosin heavy chain beta (encoded by *MYH7*), which is specific for slow-twitch oxidative type I fibers (Schiaffino & Reggiani, 2011). A decrease in type I fibers has been shown in two different androgen excess rodent models (Holmang *et al.*, 1990; Shen *et al.*, 2019). Impaired insulin sensitivity in the hyperandrogenic animals was associated with fewer type I fibers and increased type II fibers in skeletal muscles (Holmang *et al.*, 1990; Shen *et al.*, 2019), features expected to result in reduced insulin sensitivity in this tissue as a higher proportion of type I fibers leads to better insulin responsiveness (Stuart *et al.*, 2013). Moreover, a lower proportion of type I fibres correlated with the severity of insulin resistance in subjects with the metabolic syndrome (Stuart *et al.*, 2013). Androgens appear to decrease highly oxidative and insulin-sensitive type I muscle fibers and increase glycolytic and less insulin-sensitive type II fibers in non-athletes, further promoting the development of insulin resistance. However, whether these changes in muscle morphology precede or follow the development of insulin resistance in women with PCOS is not known. Moreover, the sarcoendoplasmic reticulum calcium ATPase 2 (SERCA2a), which is expressed primarily in slow-twitch skeletal muscle, was less abundant in PCOS muscle. The SERCA2a pump transport calcium ions from the cytosol back to the sarcoplasmic reticulum after muscle contraction to keep the cytosolic Ca^{2+} concentration at a low level. Its function is closely related to muscle health and function (Xu & Van Remmen, 2021). There were three sites with lower phosphorylation in SERCA2a, suggesting that lower SERCA2a expression is not only a reflection of decreased type I fibers but also an altered function. Impaired SERCA2 function may lead to increased cytosolic Ca^{2+} concentration, which in turn can impair force production and mitochondrial function in type I fibers (Xu & Van Remmen, 2021). Although serum androgen levels are positively correlated with athletic performance in female athletes (Bermón *et al.*, 2018), skeletal muscle contraction and filament sliding pathways were downregulated in muscle biopsies from hyperandrogenic women with overweight/obesity in this study. Thus, androgens may have differential actions on female skeletal muscle function in moderately physically active subjects and female athletes.

Androgens are mainly produced in the ovaries and adrenal glands but there is also a local production in adipose tissue. In overweight/obese women with PCOS, increased AKR1C3 levels mediated increased testosterone generation from androstenedione in subcutaneous adipose tissue, which enhanced lipid storage (O'Reilly *et al.*, 2017). Higher expression of AKR1C1 and AKR1C3 in PCOS skeletal muscle in this study could increase local synthesis of dihydrotestosterone via the backdoor pathway, and increase androgenic signaling in skeletal muscle.

The expression of proteins involved in lipid transport and lipid metabolism was increased in the muscle of women with PCOS. Lipid transport was clustered around apolipoprotein C1, C2 and C3. Interestingly, a recent proteomics study shows that serum apolipoprotein C3 levels are higher in insulin resistant women with PCOS compared to insulin sensitive women (Li *et al.*, 2020). Could apolipoproteins aid in the deposition of excess fat in skeletal muscle and contribute to lipotoxicity? Fatty acids in skeletal muscle that do not undergo beta-oxidation in the mitochondria inevitably

contribute to lipid synthesis. The most important single fate for these fatty acids is triacylglycerol esterification, and storage in lipid droplets. Perilipins act largely as a scaffold protein on lipid droplets. Perilipin-1 is localized in the periphery of intramuscular lipids, known as extracellular lipids (Gandolfi *et al.*, 2011 [↗](#); Zhao *et al.*, 2021 [↗](#)). In pig, muscle perilipin-1 proteins are localized around lipid droplets in mature and developing adipocytes in correspondence with extra-myocellular lipids (Gandolfi *et al.*, 2011 [↗](#); Zhao *et al.*, 2021 [↗](#)). Therefore, the high expression of perilipin-1 in muscle from women with PCOS is likely a sign of lipotoxicity. Taken together, intramuscular lipid accumulation and type I muscle fiber decline likely contribute to insulin resistance in PCOS muscle.

PCOS is associated with chronically elevated levels of inflammatory markers in the circulation (Orio *et al.*, 2005 [↗](#)) and low-grade inflammation has been suggested to play a key role in the pathophysiology (Mancini *et al.*, 2021 [↗](#)). We and others have previously shown that many of the significantly enriched gene expression pathways in PCOS muscle are involved in immune responses or are related to immune diseases in muscle (Skov *et al.*, 2007 [↗](#); Nilsson *et al.*, 2018 [↗](#)). A distinct pattern was the downregulation of a family of genes named the human leukocyte antigen (HLA) complex and downregulation of gene sets associated with inflammatory responses in the PCOS muscle (Manti *et al.*, 2020 [↗](#)). In this study, the HLA-B was the most downregulated protein, in line with lower gene expression of HLA-B in skeletal muscle (Nilsson *et al.*, 2018 [↗](#)). The HLA complex helps the immune system distinguish endogenous proteins from proteins made by foreign invaders by making peptide-presenting proteins that are present on the cell surface. When the immune system recognizes the peptides as foreign, this will trigger self-destruction of the infected cell. Numerous HLA alleles have been identified as genetic risk factors for autoimmune thyroid disease, which is increased nearly threefold in women with PCOS (Zeber-Lubecka & Hennig, 2021 [↗](#)). The co-occurrence of PCOS and autoimmune thyroid disease has led to the suggestion that PCOS itself may be an autoimmune disorder. At the same time, five different immunoglobulins were upregulated, which contributed to the enrichment of immune system pathways.

In adipose tissue, the difference between groups was small with relatively few differently expressed proteins but several of the proteins with a higher expression play a role in immune system processes including, immunoglobulins and HLA-B. Sequestosome-1 is an immunometabolic protein that is both involved in the activation of nuclear factor kappa-B and regulates mitochondrial functionality by modulating the expression of genes underlying mitochondrial respiration (Poon *et al.*, 2021 [↗](#)). Alterations in the immune response and immunometabolic pathways are seen in both muscle and adipose tissue, but whether and how these alterations affect, and potentially contribute to the pathophysiology of PCOS remains to be investigated.

Electrical stimulation-related proteome changes

Regular physical exercise improves insulin sensitivity and is the first-line approach to manage both reproductive and metabolic disturbances in those with PCOS and overweight or obesity, but adherence is low. Electrical stimulations inducing contraction could therefore be an alternative way to ameliorate symptoms in women with PCOS, alongside exercise. Transcriptomic changes in response to one single bout of electrical stimulation mimic the response to one bout of exercise (Benrick *et al.*, 2020 [↗](#)). Therefore, we hypothesize that long-term treatment with electrical stimulation or exercise also triggers overlapping signaling pathways. Five weeks of repeated treatment with electrical stimulation did not alter mRNA expression but increased the expression of several hundred proteins in skeletal muscle and about 50 proteins in adipose tissue. The most pronounced changes in both tissues were mechanisms involved in wound healing such as the increase in ECM formation and enriched pathways for nitric oxide metabolic process and Fc-gamma receptor signaling pathways. Moreover, wound healing and blood coagulation pathways were upregulated in skeletal muscle, suggesting that repeated needling induces small damages in the tissues that trigger a wound healing response. However, increased expression of ECM-related structural components such as collagens and integrins are not only involved in wound healing, but

are also increased in response to muscle contractions as previously shown in response to eccentric exercise and repeated electrical stimulation with electrodes (Mackey *et al.*, 2011 [↗](#); Hyldahl *et al.*, 2015 [↗](#)). We propose that these changes are induced by contraction but we cannot rule out that the ECM-related changes occurred as a direct result of repeated needle insertion. The delayed synthesis of collagens and subsequent strengthening of the ECM structural matrix in response to both exercise and electrical stimulation without needle insertion (Mackey *et al.*, 2011 [↗](#); Hyldahl *et al.*, 2015 [↗](#)), supports the idea that contractions induce remodeling of the ECM that may provide protective adaptation to repeated contractions.

We hypothesized that electrical stimulations mimic the response to repeated exercise. Therefore, we searched for overlap between changes in the proteome following electrical stimulation treatment in this study and a meta-analysis on the exercise-induced proteome in skeletal muscle (Padrao *et al.*, 2016 [↗](#)). Seven proteins were changed in both conditions; Fibrinogen beta chain, actin, vimentin, annexin A2, moesin, gelsolin, and hypoxanthine-guanine phosphoribosyl-transferase. Actin, vimentin and moesin make up structural parts of the cytoskeleton and all proteins are enriched in the GO terms immune system process (GO:0002376) and positive regulation of metabolic processes (GO:0009893). Current evidence suggests that there are immunometabolic alterations in skeletal muscle from women with PCOS (Skov *et al.*, 2007 [↗](#); Nilsson *et al.*, 2018 [↗](#); Manti *et al.*, 2020 [↗](#); Stepto *et al.*, 2020 [↗](#)), and increased expression of the abovementioned proteins could cause contraction-induced changes in the immunometabolic response.

Repeated treatment with electrical stimulation improves glucose homeostasis and lowers Hba1c in women with PCOS (Stener-Victorin *et al.*, 2016 [↗](#)). A not so well characterized, upregulated pathway involves exocytosis and protein transport, which is of interest with regards to glucose transporter 4 (GLUT4) translocation stimulated by AMP kinase (AMPK). The activity of AMPK in muscle increases substantially during contraction and increases glucose transport. AMPK has been linked to at least two mechanisms for the control of vesicle trafficking, namely the regulation of Rab and the generation of phosphatidylinositol 3,5-bisphosphate in the control of GLUT4 translocation (Sylow *et al.*, 2017 [↗](#)). The Rab GTPase-activating proteins TBC1D1 and TBC1D4 are thought to mediate the effects of AMPK on GLUT4 translocation and glucose transport. At present, it is unclear which specific Rab proteins might be regulated by AMPK downstream of TBC1D1 and TBC1D4, but Rab-13 appears to act downstream of TBC1D4 (Sylow *et al.*, 2017 [↗](#)). Six Rab proteins showed higher expression after electrical stimulation, including Rab-13, making these proteins potential candidates for regulating glucose uptake by electrically stimulated contractions. The proteome in adipose tissue and skeletal muscle did not show differential expression of proteins with well-known metabolic effects that can easily explain the improvement in Hba1c. This could be due to the time-point when the biopsies were collected, which was within 48 hours of the last treatment when the acute effects of electrical stimulation are no longer present and the long-term changes may be more subtle.

Limitations

The interpretation of the protein expression in this study is limited by the relatively small number of women (n =10/group), the inclusion of only Caucasian women, and a mix of PCOS phenotypes since we used the Rotterdam diagnostic criteria. Moreover, we cannot distinguish whether the identified dysregulated pathways are responses to PCOS or causal effectors. Some may argue that another limitation is the fact that we cannot identify the changes in protein expression in specific cell types i.e. adipocytes and myocytes, as the biopsies consist of many different cell types and structures, e.g. nerves, immune cells, vessels and connective tissue. However, this can also be seen as a strength as no cell acts independently of the cells surrounding it.

Conclusions

Our findings suggest that highly oxidative and insulin-sensitive type I muscle fibers are decreased in PCOS which in combination with more extramyocellular lipids, may be key factors for insulin resistance in PCOS muscle. In adipose tissue, the difference between groups was small. A 5-week treatment with electrical stimulation triggered a wound healing response in both adipose tissue and skeletal muscle. In addition, remodeling of the ECM can provide protective adaptation to repeated skeletal muscle contractions.

Data Availability

The proteomics data have been deposited to the ProteomeXchange Consortium via the Proteomics Identifications (PRIDE) (RRID:SCR_003411) partner repository with the dataset identifier PXD025358 and will be made available when the manuscript is accepted for publication. The rest of the data that support the findings of this study are available from the corresponding author upon reasonable request.

Additional information

Data availability statement

The proteomics data have been deposited to the ProteomeXchange Consortium via the Proteomics Identifications (PRIDE) (RRID:SCR_003411) partner repository with the dataset identifier PXD025358 and will be made available when the manuscript is accepted for publication. The rest of the data that support the findings of this study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare that they have no competing interests.

Author contributions

E.SV., J.N., C.L. and A.B. were involved in the conception and design of the present study, G.E., M.MS., C.P., R.B., E.N., B.J., V.E., J.L. and A.B. contributed with data acquisition and analysis, E.SV. and A.B. contributed with interpretation of data, and A.B., E.SV. G.E. and E.N. drafted the work. All authors revised the work critically for intellectual content and approved the final version of the manuscript. All authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All persons designated as authors qualify for authorship, and those who qualify for authorship are listed.

Funding

A.B. holds funding from the Swedish Research Council (2020-02485), E.SV. holds funding from the Swedish Research Council (2022-00550), the Novo Nordisk Foundation (NNF22OC0072904), and I.W.A. holds funding from the Swedish Research Council (2020-01463), Mary von Sydow Foundation and Diabetes Wellness Sverige, and JN holds funding from IngaBritt and Arne Lundberg Research Foundation. The funding bodies did not have a role in study design and have no role in the implementation of the study.

Acknowledgements

For proteomic analysis we thank Britt-Marie Olsson, Johannes Fuchs and Carina Sihlbom at the Proteomics Core Facility, and for the use of its technical equipment and support we thank the Genomics Core Facility, both at Sahlgrenska Academy, University of Gothenburg, Sweden. We are grateful to the Inga-Britt and Arne Lundberg Research Foundation for the donation of the Orbitrap Fusion Tribrid MS instrument. For analysis of array data we thank Michaela Martis at University of Linköping.

Supporting information

Additional supporting information can be found online in the Supporting Information section at the end of the HTML view of the article. Supporting information files available:

Supplemental method

Detailed description of the sample preparation and LC-MS/MS analysis.

Table S1. Differentially expressed proteins in skeletal muscle from women with PCOS compared with controls (n=10/group). $p < 0.05$, log₂ fold-change ± 0.5 .

Table S2. Gene ontology pathway analysis of differentially expressed proteins in skeletal muscle from women with PCOS compared with controls.

Table S3. Overlap between differentially expressed proteins in skeletal muscle from women with PCOS compared with controls and differentially expressed genes.

Table S4. Phosphorylation sites with a change in phosphorylation in skeletal muscle from women with PCOS compared with controls.

Table S5. Differentially expressed proteins in adipose tissue from women with PCOS compared with controls.

Table S6. Phosphorylation sites with a change in phosphorylation in adipose tissue from women with PCOS compared with controls.

Table S7. Gene expression and methylation changes in skeletal muscle after treatment with electrical stimulation.

Table S8. Changed proteins in skeletal muscle from women with PCOS after treatment with electrical stimulation.

Table S9. Gene ontology pathway analysis of changed proteins in skeletal muscle from women with PCOS after treatment with electrical stimulation.

Table S10. Phosphorylation sites with a change in phosphorylation in skeletal muscle from women with PCOS after treatment with electrical stimulation.

Table S11. Changed proteins in adipose tissue from women with PCOS after treatment with electrical stimulation.

Table S12. Gene ontology pathway analysis of changed proteins in adipose tissue from women with PCOS after treatment with electrical stimulation.

Table S13. Phosphorylation sites with a change in phosphorylation in adipose tissue from women with PCOS after treatment with electrical stimulation.

References

- Abdulkhalikova D, Sustarsic A, Vrtacnik Bokal E, Jancar N, Jensterle M, Burnik Papler T. (2022) **The Lifestyle Modifications and Endometrial Proteome Changes of Women With Polycystic Ovary Syndrome and Obesity** *Front Endocrinol (Lausanne)* **13**
- Barber TM, Franks S. (2021) **Obesity and polycystic ovary syndrome** *Clin Endocrinol (Oxf)*
- Benrick A, Pillon NJ, Nilsson E, Lindgren E, Krook A, Ling C, Stener-Victorin E. (2020) **Electroacupuncture Mimics Exercise-Induced Changes in Skeletal Muscle Gene Expression in Women With Polycystic Ovary Syndrome** *The Journal of clinical endocrinology and metabolism* **105**
- Bermon S, Hirschberg AL, Kowalski J, Eklund E. (2018) **Serum androgen levels are positively correlated with athletic performance and competition results in elite female athletes** *British journal of sports medicine* **52**:1531–1532
- Brower MA, Hai Y, Jones MR, Guo X, Chen YI, Rotter JI, Krauss RM, Legro RS, Azziz R, Goodarzi MO. (2019) **Bidirectional Mendelian randomization to explore the causal relationships between body mass index and polycystic ovary syndrome** *Hum Reprod* **34**:127–136
- Corbould A, Kim YB, Youngren JF, Pender C, Kahn BB, Lee A, Dunaif A. (2005) **Insulin resistance in the skeletal muscle of women with PCOS involves intrinsic and acquired defects in insulin signaling** *Am J Physiol Endocrinol Metab* **288**:E1047–1054
- Corton M, Botella-Carretero JI, Lopez JA, Camafeita E, San Millan JL, Escobar-Morreale HF, Peral B. (2008) **Proteomic analysis of human omental adipose tissue in the polycystic ovary syndrome using two-dimensional difference gel electrophoresis and mass spectrometry** *Hum Reprod* **23**:651–661
- Dapas M, Lin FTJ, Nadkarni GN, Sisk R, Legro RS, Urbanek M, Hayes MG, Dunaif A. (2020) **Distinct subtypes of polycystic ovary syndrome with novel genetic associations: An unsupervised, phenotypic clustering analysis** *PLoS Med* **17**
- Deutsch EW *et al.* (2020) **The ProteomeXchange consortium in 2020: enabling ‘big data’ approaches in proteomics** *Nucleic Acids Res* **48**:D1145–D1152
- Dumesic DA, Padmanabhan V, Chazenbalk GD, Abbott DH. (2022) **Polycystic ovary syndrome as a plausible evolutionary outcome of metabolic adaptation** *Reprod Biol Endocrinol* **20**
- Gandolfi G, Mazzoni M, Zambonelli P, Lalatta-Costerbosa G, Tronca A, Russo V, Davoli R. (2011) **Perilipin 1 and perilipin 2 protein localization and gene expression study in skeletal muscles of European cross-breed pigs with different intramuscular fat contents** *Meat Sci* **88**:631–637
- Group. REA-SPCW (2004) **The Rotterdam ESHRE/ASRM-sponsored PCOS Consensus Workshop Group 2004 Revised 2003 consensus on diagnostic criteria and long-term health risks related to polycystic ovary syndrome (PCOS)** *Fertil Steril* **81**:19–25

- Guo X, Puttabyatappa M, Domino SE, Padmanabhan V. (2020) **Developmental programming: Prenatal testosterone-induced changes in epigenetic modulators and gene expression in metabolic tissues of female sheep** *Mol Cell Endocrinol* **514**
- Holmang A, Svedberg J, Jennische E, Bjorntorp P. (1990) **Effects of testosterone on muscle insulin sensitivity and morphology in female rats** *Am J Physiol* **259**:E555–560
- Hutchison SK, Stepto NK, Harrison CL, Moran LJ, Strauss BJ, Teede HJ. (2011) **Effects of exercise on insulin resistance and body composition in overweight and obese women with and without polycystic ovary syndrome** *The Journal of clinical endocrinology and metabolism* **96**:E48–56
- Hyldahl RD, Nelson B, Xin L, Welling T, Groscost L, Hubal MJ, Chipkin S, Clarkson PM, Parcell AC. (2015) **Extracellular matrix remodeling and its contribution to protective adaptation following lengthening contractions in human muscle** *FASEB journal : official publication of the Federation of American Societies for Experimental Biology* **29**:2894–2904
- Insenser M, Montes-Nieto R, Martinez-Garcia MA, Escobar-Morreale HF. (2016) **A nontargeted study of muscle proteome in severely obese women with androgen excess compared with severely obese men and nonhyperandrogenic women** *Eur J Endocrinol* **174**:389–398
- Jedel E, Labrie F, Oden A, Holm G, Nilsson L, Janson PO, Lind AK, Ohlsson C, Stener-Victorin E. (2011) **Impact of electro-acupuncture and physical exercise on hyperandrogenism and oligo/amenorrhea in women with polycystic ovary syndrome: a randomized controlled trial** *Am J Physiol Endocrinol Metab* **300**:E37–45
- Joham AE, Norman RJ, Stener-Victorin E, Legro RS, Franks S, Moran LJ, Boyle J, Teede HJ. (2022) **Polycystic ovary syndrome** *Lancet Diabetes Endocrinol* **10**:668–680
- Johansson J, Redman L, Veldhuis PP, Sazonova A, Labrie F, Holm G, Johannsson G, Stener-Victorin E. (2013) **Acupuncture for ovulation induction in polycystic ovary syndrome: a randomized controlled trial** *Am J Physiol Endocrinol Metab* **304**:E934–943
- Johnson WE L. C, Rabinovic A. (2007) **Adjusting batch effects in microarray expression data using empirical Bayes methods** *Biostatistics* **8**:118–127
- Kakoly NS, Earnest A, Teede HJ, Moran LJ, Joham AE. (2019) **The Impact of Obesity on the Incidence of Type 2 Diabetes Among Women With Polycystic Ovary Syndrome** *Diabetes Care* **42**:560–567
- Kokosar M *et al.* (2016) **Epigenetic and Transcriptional Alterations in Human Adipose Tissue of Polycystic Ovary Syndrome** *Sci Rep* **6**
- Li L *et al.* (2020) **Proteomics analysis of potential serum biomarkers for insulin resistance in patients with polycystic ovary syndrome** *Int J Mol Med* **45**:1409–1416
- Liu Q, Zhu Z, Kraft P, Deng Q, Stener-Victorin E, Jiang X. (2022) **Genomic correlation, shared loci, and causal relationship between obesity and polycystic ovary syndrome: a large-scale genome-wide cross-trait analysis** *BMC Med* **20**

Mackey AL, Brandstetter S, Schjerling P, Bojsen-Moller J, Qvortrup K, Pedersen MM, Doessing S, Kjaer M, Magnusson SP, Langberg H. (2011) **Sequenced response of extracellular matrix adhesion and fibrotic regulators after muscle damage is involved in protection against future injury in human skeletal muscle** *FASEB journal : official publication of the Federation of American Societies for Experimental Biology* **25**:1943–1959

Mancini A, Bruno C, Vergani E, d'Abate C, Giacchi E, Silvestrini A. (2021) **Oxidative Stress and Low-Grade Inflammation in Polycystic Ovary Syndrome: Controversies and New Insights** *Int J Mol Sci* **22**

Manneras-Holm L, Benrick A, Stener-Victorin E. (2014) **Gene expression in subcutaneous adipose tissue differs in women with polycystic ovary syndrome and controls matched pair-wise for age, body weight, and body mass index** *Adipocyte* **3**:190–196

Manneras-Holm L *et al.* (2011) **Adipose tissue has aberrant morphology and function in PCOS: enlarged adipocytes and low serum adiponectin, but not circulating sex steroids, are strongly associated with insulin resistance** *The Journal of clinical endocrinology and metabolism* **96**:E304–311

Manti M, Stener-Victorin E, Benrick A. (2020) **Skeletal Muscle Immunometabolism in Women With Polycystic Ovary Syndrome: A Meta-Analysis** *Front Physiol* **11**

Marcinkiewicz A, Gauthier D, Garcia A, Brasaemle DL. (2006) **The phosphorylation of serine 492 of perilipin a directs lipid droplet fragmentation and dispersion** *The Journal of biological chemistry* **281**:11901–11909

McIlvenna LC, Patten RK, McAinch AJ, Rodgers RJ, Stepto NK, Moreno-Asso A. (2021) **Transforming Growth Factor Beta 1 Alters Glucose Uptake but Not Insulin Signalling in Human Primary Myotubes From Women With and Without Polycystic Ovary Syndrome** *Front Endocrinol (Lausanne)* **12**

Montes-Nieto R, Insenser M, Martinez-Garcia MA, Escobar-Morreale HF. (2013) **A nontargeted proteomic study of the influence of androgen excess on human visceral and subcutaneous adipose tissue proteomes** *The Journal of clinical endocrinology and metabolism* **98**:E576–585

Moreno-Asso A, Altintas A, McIlvenna LC, Patten RK, Botella J, McAinch AJ, Rodgers RJ, Barres R, Stepto NK. (2021) **Non-cell autonomous mechanisms control mitochondrial gene dysregulation in polycystic ovary syndrome** *Journal of molecular endocrinology* **68**:63–76

Nilsson E, Benrick A, Kokosar M, Krook A, Lindgren E, Kallman T, Martis MM, Hojlund K, Ling C, Stener-Victorin E. (2018) **Transcriptional and Epigenetic Changes Influencing Skeletal Muscle Metabolism in Women With Polycystic Ovary Syndrome** *The Journal of clinical endocrinology and metabolism* **103**:4465–4477

O'Reilly MW *et al.* (2017) **AKR1C3-Mediated Adipose Androgen Generation Drives Lipotoxicity in Women With Polycystic Ovary Syndrome** *The Journal of clinical endocrinology and metabolism* **102**:3327–3339

Orio F *et al.* (2005) **The increase of leukocytes as a new putative marker of low-grade chronic inflammation and early cardiovascular risk in polycystic ovary syndrome** *The Journal of clinical endocrinology and metabolism* **90**:2–5

- Padrao AI, Ferreira R, Amado F, Vitorino R, Duarte JA. (2016) **Uncovering the exercise-related proteome signature in skeletal muscle** *Proteomics* **16**:816–830
- Poon A, Saini H, Sethi S, O’Sullivan GA, Plun-Favreau H, Wray S, Dawson LA, McCarthy JM. (2021) **The role of SQSTM1 (p62) in mitochondrial function and clearance in human cortical neurons** *Stem Cell Reports* **16**:1276–1289
- Raja-Khan N, Kunselman AR, Demers LM, Ewens KG, Spielman RS, Legro RS. (2010) **A variant in the fibrillin-3 gene is associated with TGF-beta and inhibin B levels in women with polycystic ovary syndrome** *Fertility and sterility* **94**:2916–2919
- Ronn T *et al.* (2015) **Impact of age, BMI and HbA1c levels on the genome-wide DNA methylation and mRNA expression patterns in human adipose tissue and identification of epigenetic biomarkers in blood** *Human molecular genetics* **24**:3792–3813
- Schiaffino S, Reggiani C. (2011) **Fiber types in mammalian skeletal muscles** *Physiol Rev* **91**:1447–1531
- Shen Q, Bi H, Yu F, Fan L, Zhu M, Jia X, Kang J. (2019) **Nontargeted metabolomic analysis of skeletal muscle in a dehydroepiandrosterone-induced mouse model of polycystic ovary syndrome** *Mol Reprod Dev* **86**:370–378
- Skov V, Glinborg D, Knudsen S, Jensen T, Kruse TA, Tan Q, Brusgaard K, Beck-Nielsen H, Hojlund K. (2007) **Reduced expression of nuclear-encoded genes involved in mitochondrial oxidative metabolism in skeletal muscle of insulin-resistant women with polycystic ovary syndrome** *Diabetes* **56**:2349–2355
- Smyth GK. (2004) **Linear models and empirical bayes methods for assessing differential expression in microarray experiments** *Stat Appl Genet Mol Biol* **3**
- Stener-Victorin E, Jedel E, Janson PO, Sverrisdottir YB. (2009) **Low-frequency electroacupuncture and physical exercise decrease high muscle sympathetic nerve activity in polycystic ovary syndrome** *Am J Physiol Regul Integr Comp Physiol* **297**:R387–395
- Stener-Victorin E *et al.* (2016) **Changes in HbA1c and circulating and adipose tissue androgen levels in overweight-obese women with polycystic ovary syndrome in response to electroacupuncture** *Obes Sci Pract* **2**:426–435
- Stepito NK *et al.* (2020) **Exercise and insulin resistance in PCOS: muscle insulin signalling and fibrosis** *Endocr Connect* **9**:346–359
- Stepito NK, Moreno-Asso A, McIlvenna LC, Walters KA, Rodgers RJ. (2019) **Molecular Mechanisms of Insulin Resistance in Polycystic Ovary Syndrome: Unraveling the Conundrum in Skeletal Muscle?** *The Journal of clinical endocrinology and metabolism* **104**:5372–5381
- Stuart CA, McCurry MP, Marino A, South MA, Howell ME, Layne AS, Ramsey MW, Stone MH. (2013) **Slow-twitch fiber proportion in skeletal muscle correlates with insulin responsiveness** *The Journal of clinical endocrinology and metabolism* **98**:2027–2036
- Sylow L, Kleinert M, Richter EA, Jensen TE. (2017) **Exercise-stimulated glucose uptake - regulation and implications for glycaemic control** *Nat Rev Endocrinol* **13**:133–148

- Tiwari N, Pasrija S, Jain S. (2019) **Randomised controlled trial to study the efficacy of exercise with and without metformin on women with polycystic ovary syndrome** *Eur J Obstet Gynecol Reprod Biol* **234**:149–154
- Villa J, Pratley RE. (2011) **Adipose tissue dysfunction in polycystic ovary syndrome** *Curr Diab Rep* **11**:179–184
- Wang W, Jiang Q, Niu Y, Ding Q, Yang X, Zheng Y, Hao J, Wei D. (2022) **Proteomics and bioinformatics analysis of follicular fluid from patients with polycystic ovary syndrome** *Front Mol Biosci* **9**
- Xu H, Van Remmen H. (2021) **The SarcoEndoplasmic Reticulum Calcium ATPase (SERCA) pump: a potential target for intervention in aging and skeletal muscle pathologies** *Skelet Muscle* **11**
- Zeber-Lubecka N, Hennig EE. (2021) **Genetic Susceptibility to Joint Occurrence of Polycystic Ovary Syndrome and Hashimoto's Thyroiditis: How Far Is Our Understanding?** *Front Immunol* **12**
- Zhang X, Smits AH, van Tilburg GB, Ovaa H, Huber W, Vermeulen M. (2018) **Proteome-wide identification of ubiquitin interactions using UbIA-MS** *Nat Protoc* **13**:530–550
- Zhao Y, Albrecht E, Li Z, Schregel J, Sciascia QL, Metges CC, Maak S. (2021) **Distinct Roles of Perilipins in the Intramuscular Deposition of Lipids in Glutamine-Supplemented, Low-, and Normal-Birth-Weight Piglets** *Front Vet Sci* **8**
- Zhao Y *et al.* (2020) **Body Mass Index and Polycystic Ovary Syndrome: A 2-Sample Bidirectional Mendelian Randomization Study** *The Journal of clinical endocrinology and metabolism* **105**

Article and author information

E Stener-Victorin

Department of Physiology and Pharmacology, Karolinska Institute, 17177 Stockholm, Sweden
ORCID iD: [0000-0002-3424-1502](https://orcid.org/0000-0002-3424-1502)

G Eriksson

Department of Physiology and Pharmacology, Karolinska Institute, 17177 Stockholm, Sweden

M Mohan Shrestha

Department of Physiology, Institute of Neuroscience and Physiology, Sahlgrenska Academy, University of Gothenburg, 40530 Gothenburg, Sweden

C Perian

Department of Physiology, Institute of Neuroscience and Physiology, Sahlgrenska Academy, University of Gothenburg, 40530 Gothenburg, Sweden

B Jude

Department of Physiology and Pharmacology, Karolinska Institute, 17177 Stockholm, Sweden

V Engman

Department of Physiology and Pharmacology, Karolinska Institute, 17177 Stockholm, Sweden

R Boi

Department of Physiology, Institute of Neuroscience and Physiology, Sahlgrenska Academy, University of Gothenburg, 40530 Gothenburg, Sweden

E Nilsson

Epigenetics and Diabetes Unit, Department of Clinical Sciences, Lund University Diabetes Centre, Lund University, Malmö, Sweden

C Ling

Epigenetics and Diabetes Unit, Department of Clinical Sciences, Lund University Diabetes Centre, Lund University, Malmö, Sweden

J Nyström

Department of Physiology, Institute of Neuroscience and Physiology, Sahlgrenska Academy, University of Gothenburg, 40530 Gothenburg, Sweden

I Wernstedt Asterholm

Department of Physiology, Institute of Neuroscience and Physiology, Sahlgrenska Academy, University of Gothenburg, 40530 Gothenburg, Sweden

J T Lanner

Department of Physiology and Pharmacology, Karolinska Institute, 17177 Stockholm, Sweden

A Benrick

Department of Physiology, Institute of Neuroscience and Physiology, Sahlgrenska Academy, University of Gothenburg, 40530 Gothenburg, Sweden, School of Health Sciences, University of Skövde, 54128 Skövde, Sweden

For correspondence: anna.benrick@gu.se

ORCID ID: [0000-0003-4616-6789](https://orcid.org/0000-0003-4616-6789)

Copyright

© 2023, Stener-Victorin et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Hongmei Wang

Institute of Zoology, Chinese Academy of Sciences, Beijing, China

Senior Editor

Wei Yan

University of California, Los Angeles, Torrance, United States of America

Reviewer #1 (Public Review):

In the manuscript, the authors tried to explore the molecular alterations of adipose tissue and skeletal muscle in PCOS by global proteomic and phosphorylation site analysis. In the study, the samples are valuable, while there are no repeats for MS and there are no functional studies for the indicted proteins, phosphorylation sites. The authors achieved their aims to some extent, but not enough.

- <https://doi.org/10.7554/eLife.87592.1.sa1>

Reviewer #2 (Public Review):

This study provides the proteomic and phosphoproteomics data for our understanding of the molecular alterations in adipose tissue and skeletal muscle from women with PCOS. This work is useful for understanding of the characteristics of PCOS, as it may provide potential targets and strategies for the future treatment of PCOS. While the manuscript presents interesting findings on omics and phenotypic research, the lack of in-depth mechanistic exploration limits its potential impact.

The study primarily presents findings from omics and phenotypic research, but fails to provide a thorough investigation into the underlying mechanisms driving the observed results. Without a thorough elucidation of the mechanistic underpinnings, the significance and novelty of the study are compromised.

- <https://doi.org/10.7554/eLife.87592.1.sa0>